GenAI Energy FactSheet

DGC GREEN

Generate text

1,14 g CO_2e = 1 web search

Summarize a big document

9.8 g CO_2e

as much as

89 emails

Generate an image

2,9 g CO_2e = 2 web searches

Generate an audio

2,1 g CO_2e = 8 minutes of video call

Generate a video

54,6 g CO_2e = 1 hour of video streaming

In 2026, the **4 biggest datacenters for AI** will require as much power as the city of **New York**.

5500 MW

- Anthropic-Amazon New Carlisle
- Meta Prometheus
- OpenAI Stargate Abilene
- xAI Colossus 2

On a global scale, AI use could drain up

731 to 1,125 million cubic meters of water per year

which equates to the annual household water usage of

6 to 10 million Canadian households.

It is projected that water consumption by data centers will double by 2030.

2025 2030

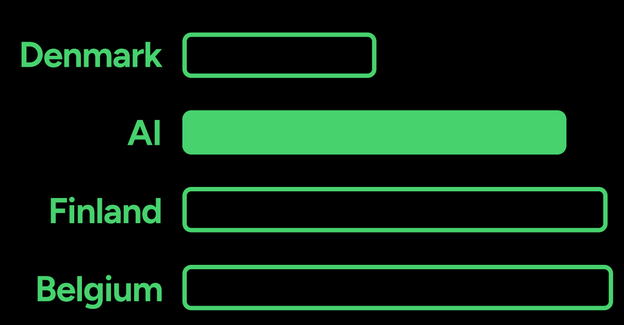
A short conversation with a chatbot uses half a litre of fresh water.

A ChatGPT session uses nearly 10x more electricity than a Google search (0.3Wh versus 2.9Wh), which is equivalent to **powering an LED light for 35 minutes.**

GOOGLE SEARCH
0.3 WH

CHATGPT SESSION
2.9 WH

In 2026, AI will be using more energy (73TWh) than Denmark and nearly as much as Finland or Belgium.
For comparison, Canada consumed 581TWh in 2024.



By 2030, AI could be using more energy (774TWh) than Brazil does today (650TWh).